# AI Collaboration and Analyst Judgment

AI collaboration and role evolution brief | Xylem / IBM SkillsBuild project

I used AI to help prepare and analyze EPA service-line inventory data and communicate the findings. For this portfolio remaster, I asked Codex to connect the research to a proposed Xylem market approach. The distinction between measured findings and proposed business action became a central part of the work.

## Data preparation and analytical choices

In the original project, AI helped interpret field names, write pandas cleaning steps, and structure checks for missing values and reported totals. Counts stored as text required comma removal and numeric conversion. Missing values were preserved rather than treated automatically as zero. The analysis retained Community Water Systems with complete required reporting in the 50 states and District of Columbia.

AI also helped examine the population denominator. Summing the EPA Population Served Count could count overlapping service populations. I chose Census resident population as a consistent state-size benchmark and labeled the resulting rates per 10,000 residents. Those rates provide context; they do not estimate affected customers.

Separating Unknown lines from Lead plus Galvanized Requiring Replacement clarified the analysis. A single total or composite score would combine unresolved classification with already-defined replacement inventory. I kept these measures separate and used their contrast to interpret state patterns.

## What I checked in the original project

My original project brief records manual review of filtering counts, duplicate Public Water System IDs, category reconciliation, terminology, and agreement between slides and notebook outputs. The final sample contained 43,993 unique systems. All category totals reconciled. In 16 records, Non-lead was missing, but the other categories already equaled the reported total; the core inventory measures were unaffected.

I also checked the calculation behind “most states” rather than relying on the scatterplot’s appearance. Unknown inventory exceeded known replacement inventory in 49 of 51 jurisdictions. Reviewing labels helped prevent Unknown lines from being described as confirmed lead and population-adjusted values from being described as affected residents.

## Communication and the portfolio remaster

AI helped organize the five-slide story, draft concise explanations, and identify crowded labels and redundant content. In the remaster, it helped develop a recommendation with traceable supporting evidence, verify published Xylem capabilities, and align the notebook, presentation and executive summary. The new proposal uses state patterns to guide utility investigation; it does not claim a completed customer assessment or pilot.

## Automated verification in the remaster

Codex reran the notebook in a fresh kernel and added checks for the new reporting-coverage calculations. The remaster generated a shared facts file and state tables for the communication artifacts. This reduced repeated transcription of numbers. These remaster checks were automated by Codex and are distinct from the original manual review described on the previous page.

Reporting coverage exposed a limitation that matters for market interpretation. New Hampshire contributes only 3 of 706 listed Community Water Systems, while coverage in the five illustrative markets ranges from 91.4% to 98.3%. Counting reporting systems does not establish how many service lines are captured. The remaster therefore keeps the observed inventories unadjusted and avoids interpreting low reported totals as proof of low need.

## Weak suggestions and corrections

An earlier AI recommendation described states as needing identification first, replacement planning, or both. That wording was too categorical. State totals cannot show that every utility has the same needs or is ready to act. The remaster treats the five states as examples, preserves both inventories, and requires utility-level evidence before selecting a service emphasis.

The original AI brief also recorded suggestions that added complexity without advancing the research question. These included a duplicate population-adjustment section and county-level analysis, although the EPA file had no county field. I kept the state scope and removed redundant analysis. The same restraint applies to the remaster: no revenue forecast, replacement-cost estimate, or pilot performance target is inferred from line counts.

Another risk was importing historical claims from a Xylem brochure. The remaster uses it to document capabilities and identifies its 2022 date. It does not reuse the brochure’s regulatory deadlines, cost figures or performance claims as current evidence. Current service scope would need confirmation before an engagement.

## How AI changes junior analyst work

AI can shorten the time spent writing routine code, formatting charts and producing first drafts. It also makes plausible but overstated conclusions easier to generate. Junior analysts still need to understand the data’s scope, choose a defensible denominator, inspect exclusions and explain what a result supports.

This project illustrates a shift toward validation and decision-focused communication. The analyst must connect findings to a useful next action while making uncertainty visible. For this project, that means recommending utility investigation based on inventory patterns and being explicit about the information still needed. Faster production is valuable only when the analysis remains reproducible and the recommendation stays within its evidence.

Evidence: original AI collaboration brief, remastered notebook and reporting-coverage table, shared facts file, and source register.